

Viscoelastic simulation and optimisation of the polymer flow through the hot-end during filament-based material extrusion additive manufacturing

Marcin P. Serdeczny, Raphaël Comminal, Md. Tusher Mollah, David B. Pedersen & Jon Spangenberg

To cite this article: Marcin P. Serdeczny, Raphaël Comminal, Md. Tusher Mollah, David B. Pedersen & Jon Spangenberg (2022) Viscoelastic simulation and optimisation of the polymer flow through the hot-end during filament-based material extrusion additive manufacturing, *Virtual and Physical Prototyping*, 17:2, 205-219, DOI: [10.1080/17452759.2022.2028522](https://doi.org/10.1080/17452759.2022.2028522)

To link to this article: <https://doi.org/10.1080/17452759.2022.2028522>



Published online: 08 Feb 2022.



Submit your article to this journal [↗](#)



View related articles [↗](#)



View Crossmark data [↗](#)



Viscoelastic simulation and optimisation of the polymer flow through the hot-end during filament-based material extrusion additive manufacturing

Marcin P. Serdeczny, Raphaël Comminal, Md. Tusher Mollah, David B. Pedersen  and Jon Spangenberg 

Department of Mechanical Engineering, Technical University of Denmark, Kgs. Lyngby, Denmark

ABSTRACT

This paper presents a novel Computational Fluid Dynamics (CFD) model that simulates the non-isothermal flow of a viscoelastic fluid through the hot-end in filament-based Material Extrusion Additive Manufacturing (MEX). The hot-end is an essential part of the printhead, as it melts and extrudes the polymeric material. Thus, robust modelling tools are necessary for optimising its operation. In this study, a viscoelastic CFD model is used to predict the filament feeding force and the results are validated against the experimental measurements at different filament feeding rates, liquefier temperatures, nozzle diameters and liquefier lengths. It is found that the viscoelastic model is more accurate than the commonly used Generalised Newtonian Fluid approximation and it captures the influence of process conditions on the feeding force. Finally, the model is used to optimise the hot-end channel. Among others, it is shown that the feeding force can be minimised by selecting an optimal liquefier diameter.

ARTICLE HISTORY

Received 11 September 2021
Accepted 10 January 2022

KEYWORDS

Fused filament fabrication;
numerical simulations;
computational fluid
dynamics; nozzle flow;
viscoelastic flow; hot-end

1. Introduction

In filament-based Material Extrusion Additive Manufacturing (MEX), also known as Fused Filament Fabrication (FFF) and Fused Deposition Modelling (FDM), the polymeric material is melted and extruded by the hot-end. The hot-end is built of a liquefier that delivers heat to the polymer and a nozzle that extrudes the molten material. At a first glance, the hot-end may seem to be a simple device, where the solid filament is turned into a molten state by the conductive heating from the liquefier walls, and it flows under the applied pressure. However, the polymer flow through the hot-end has proven difficult to predict due to the complex material rheology and physical phenomena occurring during the process (Serdeczny et al. 2020a; Mackay 2018; Serdeczny et al. 2020b). Thus, future innovations in filament-based MEX may be limited without a sound understanding and robust modelling tools of the hot-end operation.

The design of the hot-end can be optimised for specific applications such as large scale additive manufacturing, where a large throughput is desired; or printing flexible filaments, where a small feeding force is needed to avoid the filament buckling. Minimising the filament feeding force can also reduce the requirements for the stepper motor size or lower the probability of nozzle clogging due to smaller pressures and less

backflow. Furthermore, the filament-based MEX is well known for the print inaccuracies that occur at the transient states of the liquefier e.g. at the start and stop, as well as accelerations and decelerations at the corners (Comminal et al. 2019; Comminal et al. 2018). These inaccuracies are generally due to the lack of direct control over the velocity and temperature of the outflowing from the nozzle polymer. A more accurate control could be achieved if the design of the machine firmware, as well as, the toolpath generation algorithms accounted for a coupled heat and mass transfer inside the printhead.

Several analytical models have been developed to estimate the pressure drop through the printhead (Bellini et al. 2004; Mackay et al. 2017; Phan et al. 2018; Osswald et al. 2018). These models generally assume the fully developed flow of a power-law fluid. The influence of temperature on the pressure drop is accounted through a modification of rheological parameters that are valid for the entire flow domain. Thus, local temperature variations and their influence on the constitutive behaviour are not considered. Experimental measurements of the pressure inside the hot-end and the filament feeding force were performed in (Serdeczny et al. 2020a; Anderegg et al. 2019; Nienhaus et al. 2019) and compared to the analytical solutions for pressure drop. It was shown that the analytical predictions

deviate from the experiments. This was especially pronounced at the large flow rates, where the measured feeding force was observed to fluctuate (Serdeczny et al. 2020b; Nienhaus et al. 2019). Moreover, Peng et al. (2018) visualised the velocity profile inside the melt channel of the printhead and showed that it deviates from a fully developed flow profile of an isothermal power-law fluid.

Similar to other manufacturing processes, the complexity of MEX can be modelled using multi-physics Computational Fluid Dynamics (CFD) simulations (Jabbari et al. 2018). Go et al. (2017) investigated numerically the throughput limits that are related to the heat and mass exchange inside the flow channel. It was found that the maximum hot-end throughput is limited by the heat transfer and insufficient melting of the solid filament at the high flow rates. The flow and heating of an amorphous polymer were also modelled by Pigeonneau et al. (2020). It was concluded that the polymer had a perfect thermal contact with the liquefier's walls based on the comparison of the modelling results with the measured polymer temperature by Peng et al. (2018). On the other hand, Phan et al. (2018) reported that the outlet temperature of the polymer falls dramatically at high flow rates and argued that this could be due to a thermal resistance at the channel walls. Furthermore, in a CFD model presented by Serdeczny et al. (2020b), the inclusion of the thermal resistance at the channel wall improved the predictions of the filament feeding force, when compared with the measurements. In the same work, the CFD simulations were shown to capture the feeding force fluctuations that occur at the high flow rates and limit the maximum hot-end throughput. However, in some cases, discrepancies between the modelling results and experimental measurements were reported and attributed to the simplified rheological behaviour of the polymer in the numerical model.

In all of the aforementioned numerical simulations, the polymer rheology was approximated using the Generalised Newtonian Fluid (GNF) framework, where the constitutive stresses depend on the current deformation rate. In fact, the polymers are viscoelastic, which means that their internal stresses also depend on their past deformations (Bird et al. 1987), because of elastic stresses that build-up and relax during the flow. This 'memory effect' can lead to striking differences compared to the flow of GNF. In filament-based MEX, elastic stresses have been included in CFD models focusing on predicting the material deposition (Xia et al. 2019; Liu et al. 2018). However, to the knowledge of the authors, the elastic effects inside the hot-end were only included by Phan et al. (2020), where simulations

of an isothermal flow were presented and compared to the measurements of the pressure drop inside a screw-based extruder.

This work presents a novel CFD model that simulates a non-isothermal flow of a viscoelastic fluid through the hot-end in filament-based MEX. The model is validated with previously published experimental measurements of the filament feeding force at different flow rates, liquefier temperatures, nozzle diameters, and for two liquefier lengths (Serdeczny et al. 2020a). Moreover, differences in the results of the viscoelastic and GNF model are discussed. Finally the viscoelastic model is used to optimise the hot-end channel geometry by minimising the filament feeding force. Section 2 outlines the methods and materials used in the current study. Section 3 summarises and discusses the numerical results. The study is concluded in Section 4.

2. Methods and materials

2.1. Physics of the polymer flow

The flow of polymer can be described by the following conservation equations (Bird et al. 1987; QuickBooks 2020)

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot \bar{\bar{\tau}} + \rho \mathbf{g} \quad (2)$$

$$\rho C_p \frac{DT}{Dt} = k \nabla^2 T + \bar{\bar{\tau}} : \nabla \mathbf{u} \quad (3)$$

where $\mathbf{u}$ is the velocity vector; ρ is the fluid density; t is the time; p is the pressure; $\bar{\bar{\tau}}$ is the constitutive stress tensor; $\mathbf{g}$ is the gravity acceleration vector; C_p is the specific heat; T is the temperature; and k is the thermal conductivity of the polymer. In Equations (1)–(3), it is assumed that the fluid has a constant density, specific heat and thermal conductivity. Although the density of polymers is a function of temperature, its variation has a negligible influence on the flow dynamics due to a very small Reynolds number ($Re \sim 10^{-5}$) inside the hot-end. Invariance of k and C_p with respect to temperature is commonly used in the polymer processing literature (Phan et al. 2018; Osswald et al. 2018; Phan et al. 2020; Hopmann and Michaeli 2016; Xia et al. 2018). Finally, it is noted that Equation (3) does not include the heat of fusion, as an amorphous polymer without crystallinity is assumed in this work.

2.1.1. Generalised Newtonian fluid

The equations for the conservation of mass, momentum, and energy are supplemented with a constitutive model

that relates the stress tensor to the fluid deformation. If memory effects are neglected, the stress tensor solely depends on the instantaneous deformation rate, and a GNF constitutive model is used

$$\bar{\tau} = \eta \bar{S} \quad (4)$$

where η is the apparent viscosity, and $\bar{S}$ is the deformation rate tensor, calculated as $\bar{S} = (\nabla \mathbf{u} + \nabla \mathbf{u}^T)$. The apparent viscosity is modelled with the temperature-dependent power-law as

$$\eta = K(T_{ref}) \dot{\gamma}^{n-1} H(T) \quad (5)$$

where K is the consistency index at the reference temperature T_{ref} ; n is the power-law index; $\dot{\gamma}$ is the strain rate magnitude, calculated as $\dot{\gamma} = \sqrt{(\bar{S} : \bar{S})/2}$; and $H(T)$ is the shift factor function that modifies the apparent viscosity according to the polymer temperature, following the Williams-Landel-Ferry (WLF) equation

$$\log(H(T)) = -\frac{C_1(T - T_{ref})}{C_2 + (T - T_{ref})} \quad (6)$$

where C_1 and C_2 are the fitting constants.

2.1.2. Viscoelastic fluid

For more accurate modelling of the constitutive stress, accounting for memory effects (i.e. with internal stresses depending on both the current deformation rate and the history of deformations), a viscoelastic constitutive equation is used. In the viscoelastic simulations, the constitutive stress tensor is typically decomposed as

$$\bar{\tau} = \bar{\tau}_S + \bar{\tau}_P \quad (7)$$

where $\bar{\tau}_S$ is commonly called the solvent stress contribution, and $\bar{\tau}_P$ is called the polymer stress contribution. The solvent stress contribution corresponds to the instantaneous stress response and it is modelled with Newton's law

$$\bar{\tau}_S = \eta_S \bar{S} \quad (8)$$

where η_S is the solvent viscosity contribution. The polymer stress contribution is modelled with a viscoelastic constitutive equation. However, a single viscoelastic constitutive equation is unable to accurately represent the rheological data of a molten thermoplastic over the different time scales of the flow problem. Instead, the polymer stress contribution is calculated from a superposition of several viscoelastic equations, each with a different time constant. This approach is referred to as a multi-mode formulation, and the

polymer stress contribution is written as

$$\bar{\tau}_P = \sum_{i=1}^n \bar{\tau}_{P,i} \quad (9)$$

where i is the mode index, n is the number of modes, and $\bar{\tau}_{P,i}$ is the polymer stress contribution of a single mode that is calculated using the Giesekus differential viscoelastic model

$$\bar{\tau}_{P,i} + \lambda_i \check{\tau}_{P,i} + \alpha_i \frac{\lambda_i}{\eta_{P,i}} (\bar{\tau}_{P,i} \cdot \bar{\tau}_{P,i}) = \eta_{P,i} (\nabla \mathbf{u} + \nabla \mathbf{u}^T) \quad (10)$$

where λ_i , α_i , $\eta_{P,i}$ are the relaxation time, mobility factor, and polymer viscosity contribution for each mode i , respectively; $\check{\tau}_{P,i}$ is the upper-convected time derivative of $\bar{\tau}_{P,i}$ defined as

$$\check{\tau}_{P,i} = \frac{\partial \bar{\tau}_{P,i}}{\partial t} + \mathbf{u} \cdot \nabla \bar{\tau}_{P,i} - \bar{\tau}_{P,i} \cdot \nabla \mathbf{u} - \nabla \mathbf{u}^T \cdot \bar{\tau}_{P,i} \quad (11)$$

thereby making the model frame of reference invariant. The multi-mode Giesekus viscoelastic model can provide a reasonable approximation of the fluid rheology, where both the shear and elongational flows are important.

The dependency of the rheological parameters on the temperature is modelled as follows

$$\lambda_i(T) = \lambda_i(T_{ref}) \cdot H(T) \quad (12)$$

$$\eta_{P,i}(T) = \eta_{P,i}(T_{ref}) \cdot H(T) \quad (13)$$

$$\eta_S(T) = \eta_S(T_{ref}) \cdot H(T) \quad (14)$$

where $H(T)$ is the shift factor function given by Equation (6), for all the modes. Finally, it is noted that the viscoelastic models can be used for simulating a wide range of polymeric flows. They are however less often used due to the complexity of the numerical solution.

2.2. Geometry of the numerical model and boundary conditions

Figure 1 presents the domain and boundary conditions that were implemented in the numerical model. As the filament diameter D_F was slightly smaller than the liquefier diameter D_L , a short inlet section with an adiabatic perfect slip boundary condition was introduced to guide the filament inside the channel. The material was assumed to fill the entire channel (one-phase flow), including the clearance between the inserted filament and the liquefier wall. In our previous work (Serdeczny et al. 2020b), the hot-end flow was simulated without that assumption and the polymer free surface was resolved. It was shown that for most of the operating conditions, including filling of an empty liquefier (a stat-up flow) at low and moderate feeding rates, the

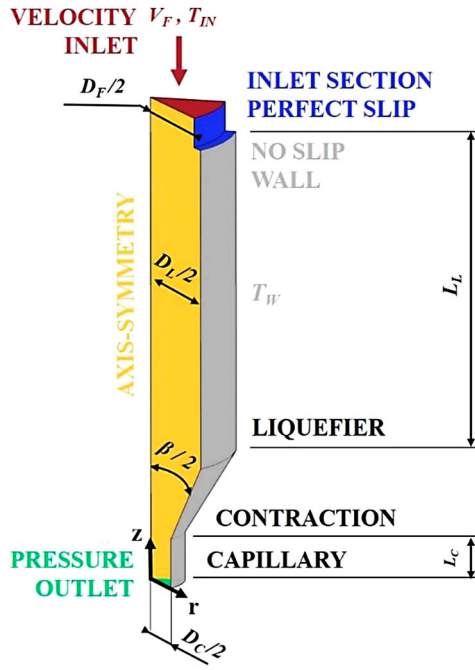


Figure 1. Geometry of the hot-end channel and boundary conditions.

polymer fills nearly the entire channel. The melt zone was shorter than the liquefier length only at very high feeding rates when the extrusion was unstable, which is typically outside of the operating conditions.

In the current model, the filament was inserted at the feeding rate V_F and inlet temperature T_{IN} into the liquefier that had a length L_L . The inlet temperature was assumed to be equal to the room temperature, as the filament is not heated above the liquefier channel and the thermal conductivity of polymers is relatively low. This assumption might be weakly violated at low feeding rates when the thermal diffusion becomes significant with respect to the advection of the material. However, the incoming filament is not in direct contact with the wall of the liquefier due to the clearance equal $\frac{D_L}{2} - \frac{D_F}{2}$. Therefore, deviation from room temperature in the filament inlet temperature is assumed to have a negligible impact on the flow through the hot-end and the predicted feeding force. This assumption however restricts our results to the 3D printers without heated chambers, where the filament is at room temperature prior to entering the printhead. After the liquefier section, the material flowed through a contraction with an angle β . The last section of the channel was a short capillary with a length L_C and diameter D_C . A zero tangential and normal velocities were set at the walls of the liquefier, contraction, and capillary sections. The wall temperature T_w was modelled with a mixed boundary condition, as explained below. The

Table 1. Dimensions of the hot-end channel used in the numerical model and experiments.

Dimension	Symbol	Unit	Value		
Filament diameter	D_F	Mm	1.75		
Liquefier diameter	D_L	Mm	2.00		
Nozzle diameter	D_C	Mm	0.25	0.40	0.80
Capillary length	L_C	Mm	0.38	0.60	1.60
Liquefier length	L_L	Mm	15.5	24.0	
Contraction angle	β	°	60		

fluid left the domain through the pressure outlet boundary that coincided with the nozzle orifice. The spatial dimensions of the hot-end channel used in both the numerical model and the experiments are summarised in Table 1. The spatial dimensions used in this study correspond to the designs that are commonly used in the commercial 3D printers.

The temperature T_w at the wall of the channel is calculated from a mixed boundary condition that accounts for the thermal resistance between the liquefier and the molten polymer (Pimenta and Alves 2020)

$$k \nabla T \cdot \mathbf{n}_w + h (T_L - T_w) = 0 \quad (15)$$

where $\mathbf{n}_w$ is the unit normal vector to the wall surface, h is the local heat transfer coefficient representing the thermal resistance at the wall of the channel, and T_L is the liquefier temperature (input to the model). The heat transfer coefficient was estimated by comparing the measured and numerically calculated filament feeding force, as discussed in Section 3.1.1. In the current model, the radiation and heat convection with air were neglected due to the one-phase flow assumption. However, the influence of those two heat transfer modes is expected to be much smaller as compared to the thermal conduction due to polymer contact with the channel walls.

2.3. Details of the numerical model

The numerical model was implemented in the open-source CFD software OpenFOAM v7.0 (OpenFOAM, Free CFD Software, The OpenFOAM Foundation, (n.d.)) with the viscoelastic library RheoTool v5.0 from Pimenta and Alves (2016). The viscoelastic solver was validated against numerical and experimental measurements in (Pimenta and Alves 2020). The simulations in this work were solved using a non-isothermal one-phase flow solver rheoHeatFoam that is a part of the RheoTool. The conservation Equations (1)–(3) were solved on a structured two-dimensional axis-symmetric mesh, which is shown in Figure 2. The two-dimensional axisymmetric mesh provided a reasonable runtime that allowed for simulating many printing conditions when compared to the three-dimensional mesh. It is

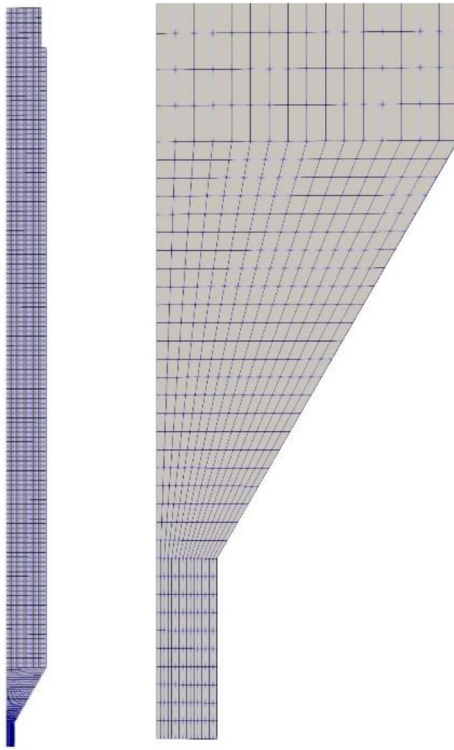


Figure 2. Example of the 2D axis-symmetric mesh used in the simulations. Left: whole domain. Right: contraction and capillary sections.

however noted that this mesh configuration enforces a symmetry around the z -axis, which might be broken at very high flow rates. Nevertheless, two-dimensional axis-symmetric mesh is expected to be a good approximation for the majority of the conditions that were simulated. The average cell length in the r -direction was $62.5 \mu\text{m}$ in the liquefier section and $25 \mu\text{m}$ in the capillary section, while in the z -direction, it was 120 and $62.5 \mu\text{m}$, respectively. The mesh was subsequently refined by decreasing the size of the elements by a half in both directions and the numerical solution was observed to be mesh independent. Thus, no further adjustments to the domain discretisation were necessary.

The solution algorithm used a segregated SIMPLEC algorithm with two correction steps on the pressure and velocity variables (Van Doormaal and Raithby 1984). The viscoelastic equations were solved using the log-conformation tensor approach that mitigates the computational instabilities (known as high Weissenberg number problem) that may occur during the viscoelastic flow simulations (Fattal and Kupferman 2004; Comminal et al. 2015; Comminal et al. 2016). The Weissenberg number relates the magnitude of the elastic forces to the viscous forces during the flow and it scales with the relaxation time and the characteristic shear rate of the flow. In the non-isothermal flows, the relaxation

time depends exponentially on the temperature (Equation (12)), and can take very large values. Therefore, the exponential increase in H (Equation (6)) with the decreasing polymer temperature was limited by setting a minimum temperature that was considered in the evolution of H to 150°C . The pressure and velocity equations were solved using the Preconditioned Conjugate Gradient (PCG) solver with the Simplified Diagonal-based Incomplete Cholesky preconditioner (DIC). The viscoelastic and temperature equations were solved using the Preconditioned Bi-conjugate Gradient (PBiCG) solver with the Simplified Diagonal-based Incomplete LU preconditioner (DILU). The choice of these solvers and preconditioners was based on the recommendations given in the RheoTool User Guide (Pimenta and Alves 2020). In our tests, they also provided the most stable and robust solution across all the simulated process conditions. More information about the solvers and preconditioners can be found in the OpenFOAM user guide (Greenshields 2018).

The transient simulations were running for 18 s . For most of the process conditions that was sufficient to reach a variation in the feeding force of less than 1% over the last 2 s , which was considered as a steady state. The steady state feeding force was of interest in order to compare the modelling results with the experimental measurements (Serdeczny et al. 2020a). The simulations were running at a Courant number of 0.05 to avoid stability and convergence issues. A relaxation factor of 0.5 was used for the pressure solver. A fully implicit scheme was used to discretise the time derivatives. The spatial gradients were discretised using a 2nd order central difference scheme. The stress divergence terms in the momentum equations were also computed using a 2nd order central difference. Moreover, the convective acceleration term in the momentum equations was neglected due to the creeping flow regime (Reynold number in order of 10^{-5}).

2.4. Post-processing of the numerical results

The numerical results were post-processed to calculate the filament feeding force and the average outlet temperature at the end of the channel. Integrating the momentum conservation (Equation (1)) over the control volume V^* shown in Figure 3 and applying the divergence theorem yields

$$\frac{D}{Dt} \int_{V^*} \rho \mathbf{u} dV = - \int_S \mathbf{p} \mathbf{n} dS + \int_S [\bar{\boldsymbol{\tau}} \cdot \mathbf{n}] dS + \int_{V^*} \rho \mathbf{g} dV \quad (16)$$

where S is the surface of volume V^* , and $\mathbf{n}$ is the unit normal vector of surface S pointing outwards the

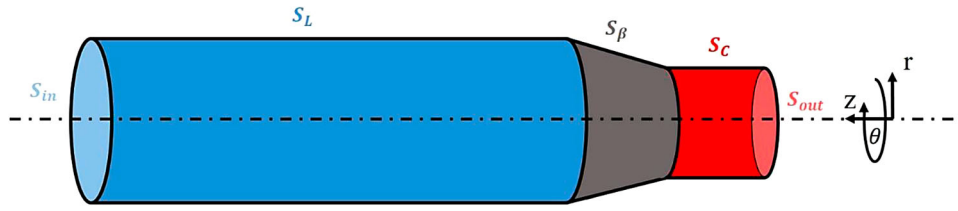


Figure 3. Control volume V^* that was used for the calculations of the filament feeding force.

control volume. As the steady state feeding force is of interest, the integral on the left-hand side of the equation becomes equal to zero. The last integral on the right-hand side that calculates the momentum due to the gravity acceleration is also neglected because of the very small volume of the channel ($\int_{V^*} \rho \mathbf{g} dV \approx 4.8 \times 10^{-4}$ N). The remaining terms from Equation (16) correspond to the static pressure and the deviatoric constitutive stresses tensor

$$\int_S (-p \mathbf{n} + \bar{\boldsymbol{\tau}} \cdot \mathbf{n}) dS = 0 \quad (17)$$

In the cylindrical coordinate system, the components of the unit normal vector and the constitutive stress tensor are: $\mathbf{n} = (n_r, n_\theta, n_z)$, where $\mathbf{n} = \sqrt{n_r^2 + n_\theta^2 + n_z^2} = 1$, and $\bar{\boldsymbol{\tau}} = \{\tau_{ij}\}$ where i and j are one of the following indices: r, θ and z . The stress component of Equation (17) in the z -direction (along which the feeding force is applied) yields

$$\int_S [\tau_{rz} n_r + \tau_{\theta z} n_\theta + (\tau_{zz} - p) n_z] dS = 0 \quad (18)$$

The surface integral in Equation (18) is computed separately for each of the surfaces limiting the control volume V^* , yielding the equation

$$F_{in} + F_{out} + F_L + F_\beta + F_C = 0 \quad (19)$$

$$F_{in} = \int_{S_{in}} (\tau_{zz} - p) dS \quad (20)$$

$$F_{out} = - \int_{S_{out}} (\tau_{zz} - p) dS \quad (21)$$

$$F_L = \int_{S_L} \tau_{rz} dS \quad (22)$$

$$F_\beta = \int_{S_\beta} \left[\tau_{rz} \cos \frac{\beta}{2} - (\tau_{zz} - p) \sin \frac{\beta}{2} \right] dS \quad (23)$$

$$F_C = \int_{S_C} \tau_{rz} dS \quad (24)$$

where $F_{in}, F_{out}, F_L, F_\beta$ and F_C are the forces acting at the surfaces of the inlet S_{in} , outlet S_{out} , liquefier S_L , contraction S_β , and capillary S_C , walls, respectively. The integrals of the components of the constitutive stress tensor and the pressure were calculated numerically during the course of the simulations. Then, the feeding force was defined as

$$F = F_{in} + F_{out} = -F_L - F_\beta - F_C \quad (25)$$

Note that F_{out} has an opposite sign to F_{in} .

The average outlet temperature of the polymer was calculated as

$$T_{out} = \frac{1}{A_{out}} \int_{S_{out}} T dS \quad (26)$$

where $A_{out} = \pi D_c^2/4$ is the area of the nozzle orifice.

2.5. Characterisation of the filament material

In this work, an ABS filament (COEX, black pigment, U.S.A.) was used for the measurements of the feeding force and the material characterisation that provided input to the numerical model.

2.5.1. Rheology

The rheology of the ABS filament was characterised through a Small Amplitude Oscillatory Shear (SAOS) test using an Anton Paar rheometer MCR 502 (Anton Paar GmbH, Austria) with a heated chamber CTD600. First, an amplitude sweep at a frequency of 10 Hz and temperature of 200 °C was performed to find the strain limit of the linear viscoelastic region. This was followed by frequency sweeps with a strain of 0.5%, at different temperatures ranging from 125 to 250 °C with an

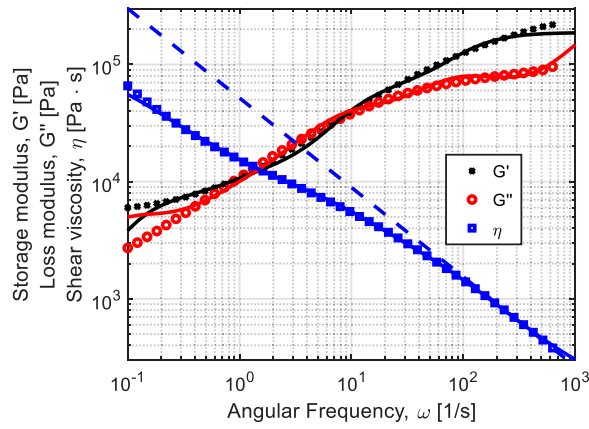


Figure 4. Rheological characterisation of ABS filament at $T_{ref} = 200$ °C. The measurements are represented by dots, the fit to the viscoelastic model by the continuous lines, while the fit to the power-law model by the dashed line.

increment of 25 °C. The SAOS test delivered information about the magnitude of the storage G' and loss G'' moduli that were used to calculate the complex viscosity. The shear viscosity was calculated from the complex viscosity with the Cox-Merz rule. The reference temperature was chosen to be 200 °C. The typical range of shear rates in the FFF process is between 10^2 and 10^3 1/s, which is typically not accessible in the oscillatory measurement. However, oscillatory measurements performed at different temperatures can be superimposed to extend the range of shear rates and create a master curve of the shear viscosity. The procedure and resultant master curve were shown in our previous publication (Serdeczny et al. 2020b). It was shown that slope of the shear viscosity for ABS was approximately constant above the shear rate of 100 1/s.

In this work, two rheological models were used: Model 1 that assumes a power-law viscosity (Equation (5)), formulated within the GNF framework; and Model 2 that includes a viscoelastic constitutive stress (Equations (7)–

(11)). The measured shear viscosity was fitted with the power-law model at the reference temperature using MATLAB. Moreover, the storage modulus, loss modulus, and shear viscosity were fitted to a four-mode Giesekus constitutive model (Equations (7)–(11)) using the Polymat software, which is an automatic fitting tool that is a part of the ANSYS Polyflow package. The measured and fitted curves are shown in Figure 4, while the fitting constants are provided in Table 2. It is noted that the power-law index is slightly lower than usually seen in the literature (Mackay et al. 2017), however, no drop in the shear stress was observed during the measurement that would be a sign of polymer slip on the surface of the measuring cone. The frequency sweeps performed at different temperatures were used to obtain the fitting constants in the WLF model (Equation (6)). The best fitting values are given in Table 2.

2.5.2. Differential Scanning Calorimetry

A Differential Scanning Calorimetry (DSC) was performed using a DSC Q200 (TA Instruments, USA) to find the specific heat of the ABS polymer, as well as to identify any crystallisation or melting peaks. A sample of 9 mg was heated from –80 to 250 °C with an average heating rate of 3 °C/min using an aluminium pan and nitrogen as an inert gas. The heat flow curve is shown in Figure 12 in the Supplemental Information. The glass transition temperature was identified in the heat flow curve at around 105 °C, and no crystallisation or melting peaks were found, as expected for the amorphous polymer. The heat flow measured by the instrument dq/dt can be related to the specific heat capacity of the material C_p as

$$\frac{dq}{dt} = MC_p \frac{dT}{dt} \quad (27)$$

where M is the mass of the sample. Within the measured temperature range, the specific heat varied from around

Table 2. Rheological parameters of the ABS filament.

Model 1 – GNF						
Variable	Symbol	Unit	Value			
Consistency index	K_{ref}	$\text{Pa} \cdot \text{s}^n$	51750			
Power-law index	n	–	0.24			
Model 2 – Viscoelastic fluid						
Variable	Symbol	Unit	Value			
			Mode 1	Mode 2	Mode 3	Mode 4
Relaxation time	λ	s	0.01	0.1	1	10
Polymer viscosity contribution	η_P	$\text{Pa} \cdot \text{s}$	1212	5322	5392	75 368
Mobility factor	α	–	0.52	0.55	0.59	0.68
Solvent viscosity contribution	η_S	$\text{Pa} \cdot \text{s}$	135			
WLF model						
Variable	Symbol	Unit	Value			
WLF Constant 1	C_1	–	3.8			
WLF Constant 2	C_2	°C	155.2			
Reference temperature	T_{ref}	°C	200			

Table 3. Other material parameters used in the numerical simulations.

Material parameter	Symbol	Unit	Value	Source
Specific heat	C_p	J/(kgK)	1500	Measured
Glass temperature	T_g	°C	105	Measured
Heat of fusion	Q_f	J	0	Measured
Thermal conductivity	k	W/(mK)	0.33	(Osswald et al. 2018)
Density	ρ	kg/m ³	945	(Osswald et al. 2018)

1000 J/(kgK) to 2000 J/(kgK), and an average value of 1500 J/(kgK) was used in the numerical model. The material parameters that were used in the simulations are summarised in Table 3.

2.6. Filament feeding force measurements

In this section, we briefly outline the experimental measurements that were used to validate the numerical model. It is noted however that the measurement results contained in this article are not novel and were previously published in (Serdeczny et al. 2020a), where more details can be found.

In the experimental setup of (Serdeczny et al. 2020a), the filament feeding force was measured using a load cell mounted to the hot-end on a specially designed testing bench. For each process condition (combination of V_F , T_L , D_C , L_L), the force was measured for one minute, where the first half a minute was disregarded due to transient conditions. For the second half, an average feeding force and a standard deviation were calculated, where the latter indicated the fluctuations of the feeding force with respect to time. For each process conditions an average from three repetitions was calculated. The experimental uncertainty of the feeding force measurement was found to be 0.8 N.

The measurement conditions and hot-end dimensions were chosen based on the commonly used parameters during the FFF process. The force was measured for different feeding rates, starting at 40 mm/min and up to a point, where the extrusion failed due to grinding of the filament by the extruder gear wheels. Moreover, three different liquefier temperatures were tested i.e. 200, 225, 250 °C.

3. Results and discussions

3.1. Validation of the numerical model

This section focuses on the evaluation of the performance of the numerical model by comparing the results to the experimental measurements of the filament

feeding force (Serdeczny et al. 2020a) that were outlined in Section 2.6.

3.1.1. Estimation of the thermal resistance at the channel wall

Following the work of Phan et al. (2018), it is suspected that the thermal resistance at the channel wall significantly reduces the wall temperature to which the polymer is exposed. Thus, the unknown wall temperature is considered one of the main sources of the uncertainty in the current computational model. In this section, the thermal resistance at the channel wall, represented by the heat transfer coefficient, is varied until the best agreement between the computed and measured filament feeding force is obtained. Figure 5 shows the feeding force (top) and outlet temperature of the polymer (bottom) versus feeding rates, simulated with the Model 1 – GNF (Figure 5(a,c)), and with Model 2 – viscoelastic fluid (Figure 5(b,d)). First, a large heat transfer coefficient is assumed ($h = 10^6$ W/m²K), which simulates no thermal resistance between the channel wall and the polymer. Model 1 has a good agreement with experiments at the low filament feeding rates (V_F), but it quickly deviates from the measurements when V_F is increased. For Model 2, the agreement with experiments also holds for the moderate feeding rates and discrepancies appear only at the high feeding rates. Interestingly, in the lower range of feeding rates, the temperature at the outlet increases when the feeding rate is increased, which is due to viscous heating of the polymer. It is noted that the viscous heating effect is larger for the viscoelastic fluid than for the GNF which is attributed to the presence of the elastic stresses next to the viscous stresses in the viscoelastic model. In the next step, the heat transfer coefficient is reduced to $h = 4000$ W/(m²K). Now, the predicted feeding force becomes closer to the measured values at the high feeding rates, but Model 1 over-predicts the measurements at the low feeding rates. Finally, the heat transfer coefficient is reduced to $h = 1000$ W/(m²K). For Model 1, this leads to a significant over-prediction of the feeding force. For Model 2, the feeding force is slightly overestimated at the low feeding rates but the slope of the curve is closer to the measured one. Moreover, it is observed that the outlet temperature of the polymer drops below the liquefier setpoint ($T_L = 200$ °C) at the highest simulated flow rate (see the bottom graphs of Figure 5). This temperature drop is however smaller than expected from the measurements reported in Phan et al. (2018), where the outlet temperature was up to 50°C below the liquefier temperature. On the other hand, our results are more consistent with the measurements of Peng

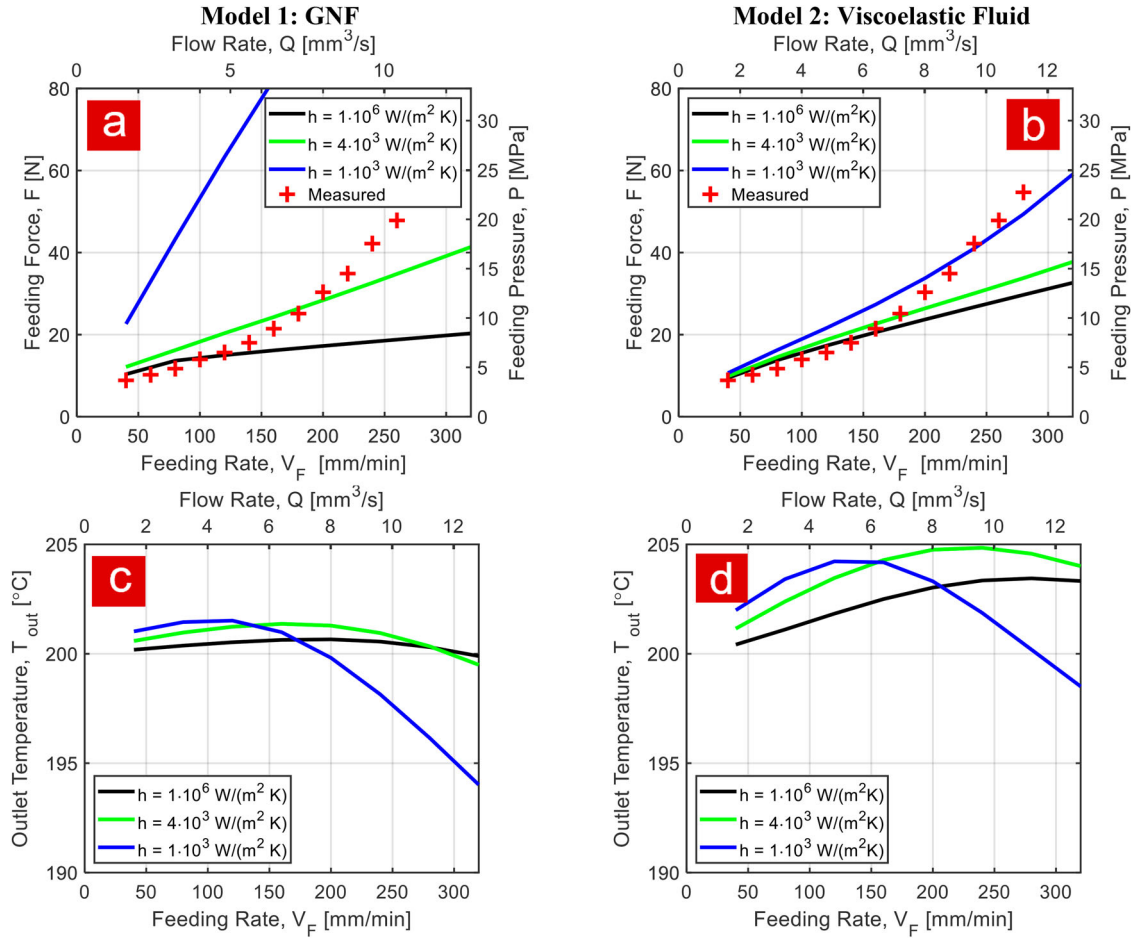


Figure 5. Effect of the heat transfer coefficient at the channel wall on the filament feeding force (top) and the outlet temperature of the polymer (bottom). Moreover, $L_L = 15.5$ mm, $D_C = 0.4$ mm and $T_L = 200$ °C.

et al. (2018) and Serdeczny et al. (2020a), where the outlet temperature was not dropping more than 20 °C below the liquefier temperature. However, a direct comparison between the measured data from (Serdeczny et al. 2020a; Peng et al. 2018) and the simulated outlet temperature was not possible due to differences in the setups. Finally, it is noted that the current model does not resolve the free surface of the fluid, which might be important in predicting the feeding force and outlet temperature at the high flow rates, where an unstable extrusion regime is observed (Serdeczny et al. 2020b). At this stage, it is difficult to clearly state which of the models is more promising. Thus, in the following section, the heat transfer coefficient is kept constant at $h = 4000$ W/(m²K) for Model 1, and $h = 1000$ W/(m²K) for Model 2, which are the values yielding the best agreement with the measurements. At the same time, it is noted that these values of the heat transfer coefficient fall within the range presented in the polymer processing literature (Liu 2014; Fradette et al. 1996).

3.1.2. Effect of the liquefier temperature

Figure 6 presents the simulated (lines) and measured (dots) filament feeding forces versus feeding rates for varying liquefier temperatures. The measurements show that the feeding force decreases with the increasing liquefier temperature, as the polymer viscosity decreases. Moreover, at high flow rates, the force increases with a larger slope, in the experiments, which indicates insufficient melting of the filament core. The inability of Model 1 to capture the force increase at the large feeding rates might be related to an inaccurate description of the extensional flow in the contraction section of the hot-end channel. In the contraction section, the polymer is stretched in the direction of the flow, causing an increase in the stored elastic energy of the fluid. The stored elastic energy has to be first generated by the additional pressure drop that is neglected in Model 1 (Hopmann and Michaeli 2016; Wang and James 2011). On the other hand, Model 2 slightly over-predicts the force at the low feeding rates

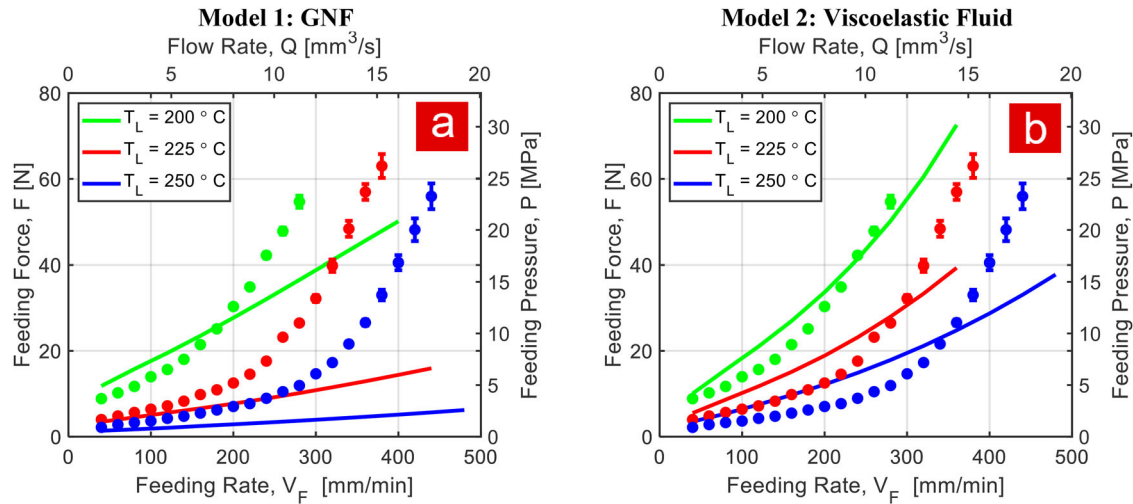


Figure 6. Validation of the simulated feeding force (lines) against experimental measurements (dots) for two models. The figure also captures the influence of the liquefier temperature T_L on the agreement of the results. Moreover, $L_L = 15.5$ mm and $D_C = 0.4$ mm. The error bars indicate force fluctuations during the measurement equal to $1.5 \sigma (F)$. See Serdeczny et al. (2020a) for more details.

but it provides a more accurate estimate of the force at the high feeding rates. Moreover, the slopes of the curves simulated with Model 2 are closer to the measured ones (see Figure 6). The agreement is however still not perfect and the remaining discrepancies between the numerical results of Model 2 and the measurements at high feeding rates could be due to the assumption that the polymer melt fills the entire volume of the liquefier. In fact, the length of the melt zone at high feeding rates may start to fluctuate, which influences the average feeding force (Serdeczny et al. 2020b). The overestimation of the force at low and moderate feeding rates might be due to the filament inlet temperature being higher than the room temperature which was assumed. Furthermore, the rheological characterisation of the polymer was based on small amplitude oscillatory shear measurements, which deliver information about linear viscoelasticity. Further rheological characterisation, including measurements of the extensional viscosity would increase the accuracy of constitutive modelling.

3.1.3. Effect of the channel geometry

Figure 7 presents the simulated (lines) and measured (dots) filament feeding forces versus feeding rates for different nozzle diameters (top), and two liquefier lengths (bottom). As expected, the measurements show larger force for smaller nozzle diameters, as a larger pressure drop occurs in the channels with smaller cross-sectional areas. The numerical models capture this trend, however, the change in the feeding force as a function of the nozzle diameter simulated with Model 1 is much smaller than in the measurement. The same

limitation of the GNF model was observed in (Serdeczny et al. 2020b), where the polymer flow through the hot-end was calculated accounting for the free surface of the fluid. The discrepancy is again attributed to the inaccurate approximation of the extensional flow of a polymer melt by the GNF model. The pressure drop due to the extensional flow depends on the contraction ratio between the liquefier and nozzle diameters. A more accurate prediction is obtained by Model 2, where the viscoelastic effects are included. The force is however slightly overestimated, which might come from the assumption of a totally filled liquefier by the polymer.

The predicted filament feeding force is compared against the measurements at different liquefier lengths in the bottom graphs of Figure 7. The measurements show that the feeding force is nearly the same for two liquefiers at the low feeding rates, whereas at the high feeding rates, the measured feeding force remains lower for the longer liquefier. A longer heating channel provides a larger surface area to heat the polymer before it enters the contraction section. Model 1 fails to predict this aspect, which is attributed to the inaccurate rheological model, as discussed earlier. On the other hand, Model 2 predicts the measurements qualitatively well and captures the effect of extending the liquefier length i.e. a lower feeding force in a longer liquefier at the high V_F . However, the feeding force for the longer channel is over-predicted. This may be attributed to the one-phase flow approximation that assumes that the entire liquefier is filled with polymer. In reality, the melt zone is expected to be shorter than the liquefier length, leading to smaller shear stresses in the liquefier section.

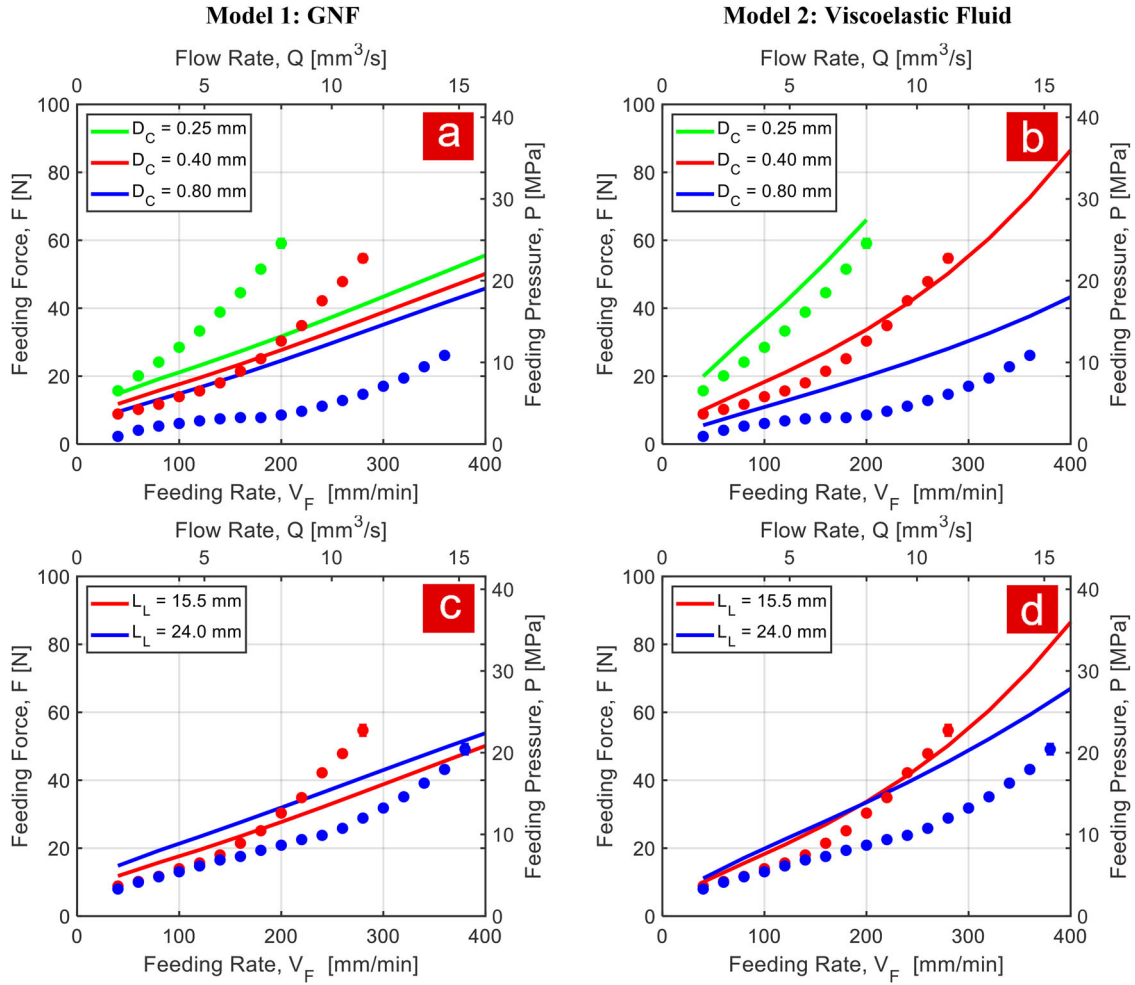


Figure 7. Validation of the simulated feeding force (lines) against experimental measurements (dots) for varying nozzle diameter (top) and liquefier lengths (bottom). The liquefier temperature was $T_L = 200$ °C.

The presented analysis showed that Model 1 can give a reasonable estimate of the feeding force at low feeding rates, but it deviates from the measurements when V_F is increased, as it neglects the elastic effects of the polymer melt that are important in the hot-end flow. Although Model 2, slightly overestimates the force in some situations, overall it gives more accurate predictions of the force in different conditions i.e. for varying feeding rate, liquefier temperature and channel geometry. Therefore, in the next section, Model 2 is used to optimise the hot-end geometry by minimising the filament feeding force.

3.2. Optimisation of the hot-end channel geometry

In this section, a parametric study of the hot-end channel dimensions is performed to investigate their influence on the filament feeding force. In the following sections, the feeding force was decomposed into

contributions from the liquefier, contraction and capillary sections, as described by Equations (22)–(25). It was found that the force contributions from the liquefier and contraction sections (F_L and F_β , respectively) were the largest, while the force coming from the wall shear stress at the capillary section (F_C) was always below 1 N. Thus, F_C will not be shown in the following graphs.

3.2.1. Liquefier length

Figure 8 presents the effect of the liquefier length on the total feeding force (Figure 8(a)) and the separate force contributions from the liquefier and contraction sections (Figure 8(b)). It is observed that the force increases abruptly if the liquefier length is too short for a given feeding rate. The increase in the feeding force is caused by an increase in the force contribution from the contraction section (see Figure 8; right). The extensional flow in the contraction section causes a large pressure drop if the liquefier length is insufficient to

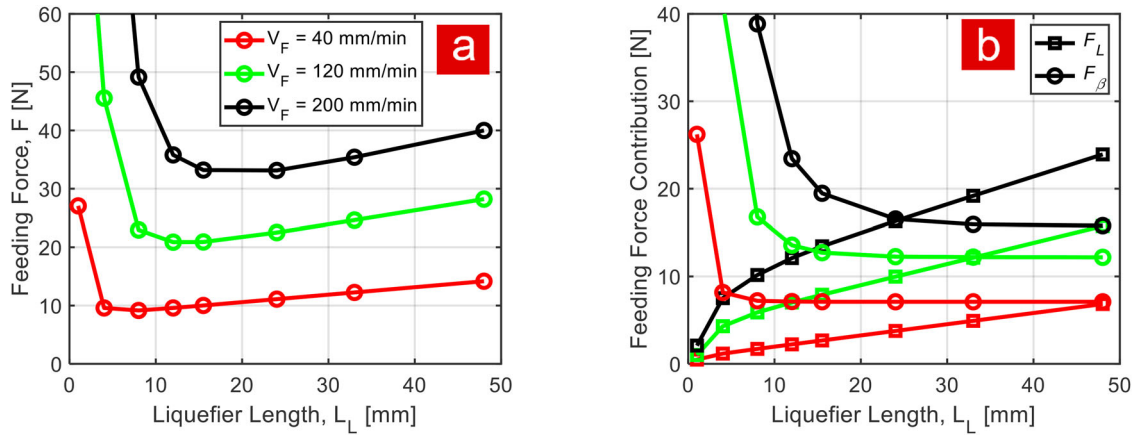


Figure 8. Effect of the liquefier length on the feeding force (a); and force contributions from the liquefier and contraction sections (b). Simulated for ABS polymer at $T_L = 200$ °C and $D_C = 0.4$ mm.

fully melt the polymer before it reaches the contraction. On the other hand, in the right plot, it can be seen that increasing the liquefier length, leads to an increase in the force contribution from the liquefier section, as a longer channel relates to a larger pressure drop. However, the increase in F_L is mitigated by the decrease in F_β , which shows that the liquefier channel can be extended to increase the maximum feeding rate without a significant increase in the feeding force.

3.2.2. Capillary length

Figure 9 presents the impact of the capillary length on the total feeding force (Figure 9(a)) and the separate force contributions from the liquefier and contraction sections (Figure 9(b)). It can be seen that the increase in the capillary length leads to a substantial increase in the feeding force. The increase in the feeding force is due to the force contribution from the contraction section. Although the force coming from the shear

stress integration at the capillary section (see Equation (24)) is relatively small, the pressure drop through the capillary section (proportional to the capillary length) increases the pressure applied on the tapered surface of the contraction section, leading to larger F_β (see Equation (23)). This numerical result suggests that the capillary section should be designed as short as possible in order to minimise the filament feeding force. On the other hand, a certain capillary length is required to stabilise the flow before the deposition. Otherwise, the extruded strand might be inconsistent due to possible instabilities induced by the viscoelastic effects in the contraction section. The significance of the capillary length for the hot-end design was also pointed by Phan et al. (2020).

3.2.3. Liquefier diameter

Figure 10 shows the dependency of the total feeding force (Figure 10(a)) and the separate force contributions

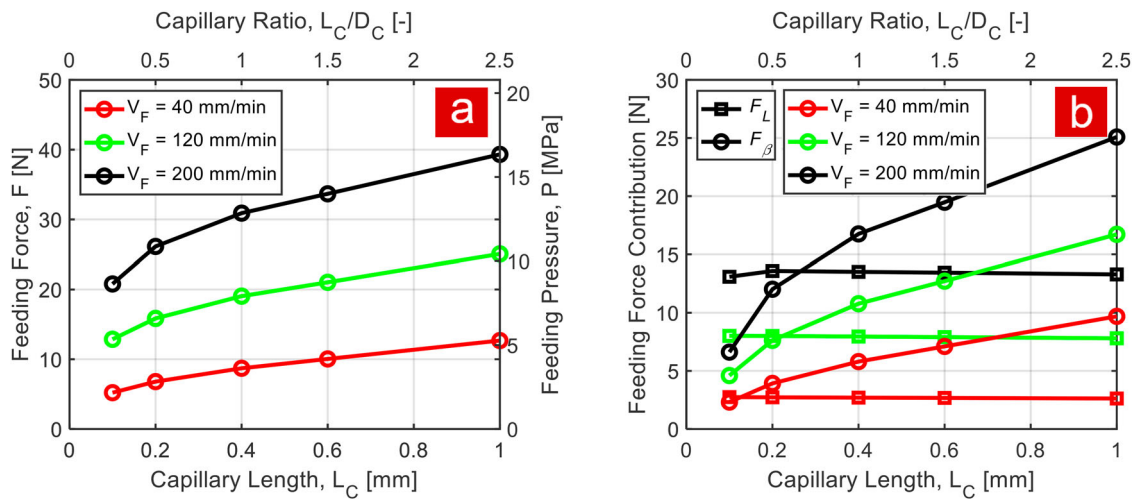


Figure 9. Effect of the capillary length on the feeding force (a), and force contributions from the liquefier and contraction sections (b). Simulated for ABS at $T_L = 200$ °C, $L_L = 15.5$ mm, and $D_C = 0.4$ mm.

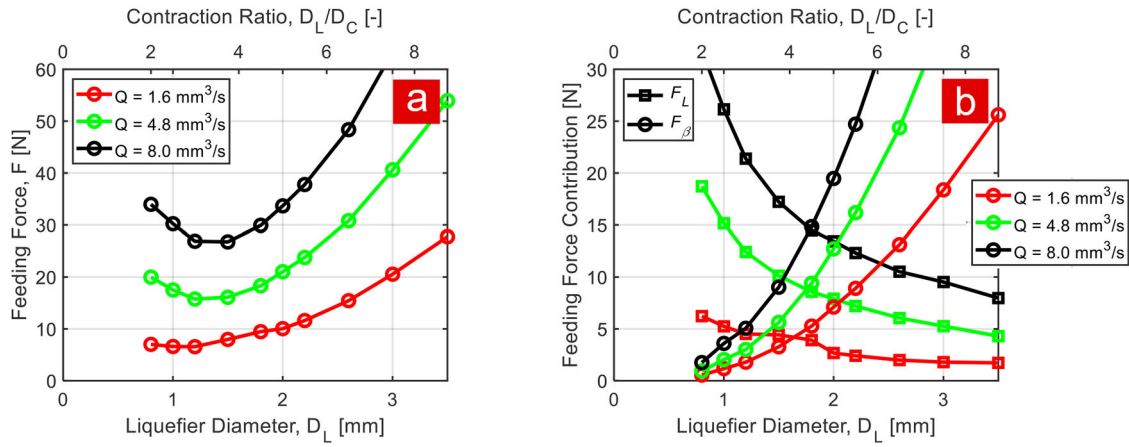


Figure 10. Effect of the liquefier diameter on the feeding force (a), and force contributions from the liquefier and contraction sections (b).

(Figure 10(b)) on the liquefier diameter. In these simulations, the ratio of the filament diameter to the liquefier diameter was kept constant and equal to $D_F/D_L = 0.875$, as used in a standard hot-end. Moreover, the nozzle diameter D_C , the contraction angle β , as well as the total length of the hot-end channel were kept constant. The study was performed for three values of the flow rate Q , which for filament diameter $D_F = 1.75 \text{ mm}$ are equivalent to the feeding rates 40, 120, and 200 mm/min (as used in the previous sections). It can be seen that the dependency of the feeding force on the liquefier diameter is non-monotonic. First, F decreases due to the decreasing force contribution from the liquefier section (see the right graph). The force contribution F_L decreases when the liquefier diameter is increased, as the average flow velocity and shear rate become smaller. On the other hand, F_β increases with D_L , due to the larger extensional rate in the contraction section (which is a function of the contraction ratio

D_L/D_C). Hence, for moderate and high flow rates, a minimum in the feeding force can be seen at around $D_L = 1.5 \text{ mm}$ (requiring a non-standard filament diameter of 1.31 mm). In this study, ABS filament was used as a model polymer, but the optimal barrel diameter may vary for materials with different rheological behaviour. This results shows that liquefiers designed for a filament diameter 2.85 mm may require more than twice the filament feeding force of standard liquefiers with a filament diameter 1.75 mm (using a standard nozzle diameter $D_C = 0.4 \text{ mm}$, in both cases).

3.2.4. Contraction angle

Figure 11 presents the influence of the contraction angle β on the total feeding force (Figure 11(a)) and the separate force contributions (Figure 11(b)). It is observed that increasing the contraction angle β leads to a decrease in the feeding force due to a lower force contribution at the contraction section (right graph). Although the

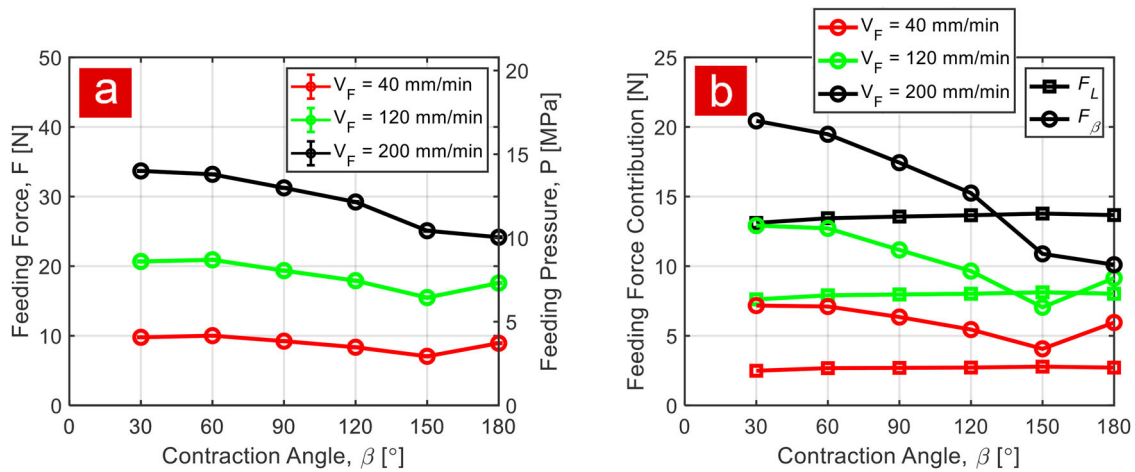


Figure 11. Influence of the contraction angle on the feeding force (a), and force contributions from the liquefier and contraction sections (b).

force is lower at increased contraction angles, a large recirculation vortex was observed to form at $\beta \geq 120^\circ$, that is generally undesired in the polymer processing (Bird et al. 1987). Moreover, at $\beta = 180^\circ$, the force contribution from the contraction section (and the total feeding force) oscillated in time. Thus, an optimum contraction angle around 90° was found to minimise the feeding force while preventing the formation of recirculation vortices.

4. Conclusions

This work presented a state-of-the-art CFD model of the polymer flow inside the printhead channel during filament-based MEX. Non-isothermal viscoelastic flow simulations were performed and compared to a simpler modelling approach that utilises the GNF framework. The numerical model was calibrated against the experimental measurements of the filament feeding force by introducing the thermal resistance at the channel wall. It was shown that both approaches were able to model accurately the filament feeding force at low feeding rates. However, when the feeding rate was increased, the viscoelastic model provided a more accurate prediction of the force than GNF model. Moreover, the viscoelastic model captured qualitatively well the dependency of the feeding force on the liquefier temperature, liquefier length, and nozzle diameter. The advantage of the viscoelastic model over the GNF model lied in the more accurate representation of the polymer rheology in the extensional flow that is present in the contraction section. However, the agreement between the experimental measurements and numerical results of the viscoelastic model was not perfect. The observed discrepancies could be due to the lack of free surface modelling that renders the actual contact area between the liquefier wall and the polymer. Moreover, the agreement could be potentially improved, if the rheological characterisation was supplemented by measurements of the extensional viscosity. Finally, it is noted that the simulations were performed using a two-dimensional axisymmetric mesh that is considered a good approximation for low and moderate feeding rates but may become invalid at high flow rates. Verifying the axisymmetric assumption at high flow rates could be a subject of future studies.

In the second part of the study, the viscoelastic model was used to optimise the channel geometry by minimising the filament feeding force. The effect of the liquefier length, capillary length, liquefier diameter and contraction angle on the flow through the hot-end were studied. It was found that extending the liquefier length increases the maximum throughput through

the hot-end by enhancing its heating power, without a significant increase in the feeding force. On the other hand, if the liquefier length is too short, the force increases abruptly due to insufficient melting of the filament core. Increasing the capillary length led to a significant increase in the feeding force due to the larger pressure drop through the capillary section. Interestingly, the dependency of the feeding force on the liquefier diameter was non-monotonic, due to opposite effects of increasing D_L on the force contributions from the liquefier and contraction sections. Thus, an optimum liquefier diameter around $D_L = 1.5$ mm was found for the investigated polymer (ABS). Increasing the contraction angle decreased the feeding force, however at large β , a recirculation vortex was observed in the contraction section and eventually led to fluctuations of the feeding force. In future works, the presented model could be used to investigate alternative geometries of the melting channel and analyse the transient responses of the flow to changes in the filament feeding rate.

Acknowledgments

The authors would like to thank Morteza Alehosseini from DTU Health Tech for the help in performing the DSC measurement.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The authors would like to acknowledge the support of the Danish Council for Independent Research (DFF), Technology and Production Sciences (FTP) (Contract No. 7017-00128 and Contract No. 8022-00042B).

ORCID

David B. Pedersen  <http://orcid.org/0000-0002-7013-7614>
Jon Spangenberg  <http://orcid.org/0000-0002-5411-6821>

References

- Anderegg, D. A., H. A. Bryant, D. C. Ruffin, S. M. Skrip, J. J. Fallon, E. L. Gilmer, and M. J. Bortner. 2019. "In-situ Monitoring of Polymer Flow Temperature and Pressure in Extrusion Based Additive Manufacturing." *Additive Manufacturing* 26: 76–83. doi:10.1016/j.addma.2019.01.002.
- Bellini, A., S. Güçeri, and M. Bertoldi. 2004. "Liquefier Dynamics in Fused Deposition." *Journal of Manufacturing Science and Engineering* 126: 237. doi:10.1115/1.1688377.
- Bird, R. B., R. C. Armstrong, and O. Hassager. 1987. *Dynamics of Polymeric Liquids*. John Wiley & Sons.

- Comminal, R., J. H. Hattel, M. A. Alves, and J. Spangenberg. 2016. "Vortex Behavior of the Oldroyd-B Fluid in the 4-1 Planar Contraction Simulated with the Streamfunction-log-Conformation Formulation." *Journal of Non-Newtonian Fluid Mechanics* 237: 1–15. doi:10.1016/j.jnnfm.2016.09.005.
- Comminal, R., M. P. Serdeczny, D. B. Pedersen, and J. Spangenberg. 2018. "Numerical Modeling of the Material Deposition and Contouring Precision in Fused Deposition Modeling", In: *Proc. Annu. Int. Solid Free. Fabr. Symp.* pp. 1855–1864.
- Comminal, R., M. P. Serdeczny, D. B. Pedersen, and J. Spangenberg. 2019. "Motion Planning and Numerical Simulation of Material Deposition at Corners in Extrusion Additive Manufacturing." *Additive Manufacturing* 29: 100753. doi:10.1016/j.addma.2019.06.005.
- Comminal, R., J. Spangenberg, and J. H. Hattel. 2015. "Robust Simulations of Viscoelastic Flows at High Weissenberg Numbers with the Streamfunction/log-Conformation Formulation." *Journal of Non-Newtonian Fluid Mechanics* 223: 37–61. doi:10.1016/j.jnnfm.2015.05.003.
- Fattal, R., and R. Kupferman. 2004. "Constitutive Laws for the Matrix-Logarithm of the Conformation Tensor." *Journal of Non-Newtonian Fluid Mechanics* 123: 281–285. doi:10.1016/j.jnnfm.2004.08.008.
- Fradette, L., P. A. Tanguy, P. Hurez, and D. Blouin. 1996. "On the Determination of Heat Transfer Coefficient Between pvc and Steel in Vacuum Extrusion Calibrators." *International Journal of Numerical Methods for Heat & Fluid Flow* 6 (1): 3–12. doi:10.1108/EUM00000000004095.
- Go, J., S. N. Schifres, A. G. Stevens, and A. J. Hart. 2017. "Rate Limits of Additive Manufacturing by Fused Filament Fabrication and Guidelines for High-Throughput System Design." *Additive Manufacturing* 16: 1–11. doi:10.1016/j.addma.2017.03.007.
- Greenshields, C. J. 2018. "OpenFOAM User Guide Version 6, OpenFOAM Found". Ltd. U-94. <https://cfd.direct/openfoam/user-guide/>.
- Hopmann, C., and W. Michaeli. 2016. *Extrusion Dies for Plastics and Rubber Design and Engineering Computations*. doi:10.3139/9781569906248.007.
- Jabbari, M., I. Baran, S. Mohanty, R. Comminal, M. R. Sonne, M. W. Nielsen, J. Spangenberg, and J. H. Hattel. 2018. "Multiphysics Modelling of Manufacturing Processes: A Review." *Advances in Mechanical Engineering* 10: 1–31. doi:10.1177/1687814018766188.
- Liu, Y. 2014. "Heat Transfer Process Between Polymer and Cavity Wall During Injection Molding", PhD diss., Universitätsverlag Chemnitz.
- Liu, J., K. L. Anderson, and N. Sridhar. 2018. "Direct Simulation of Polymer Fused Deposition Modeling (FDM) - An Implementation of the Multi-Phase Viscoelastic Solver in OpenFOAM." *International Journal of Computational Methods* 15: 1844002. doi:10.1142/S0219876218440024.
- Mackay, M. E. 2018. "The Importance of Rheological Behavior in the Additive Manufacturing Technique Material Extrusion." *Journal of Rheology* 62: 1549–1561. doi:10.1122/1.5037687.
- Mackay, M. E., Z. R. Swain, C. R. Banbury, D. D. Phan, and D. A. Edwards. 2017. "The Performance of the hot end in a Plasticating 3D Printer." *Journal of Rheology* 61: 229–236. doi:10.1122/1.4973852.
- Nienhaus, V., K. Smith, D. Spiehl, and E. Dörsam. 2019. "Investigations on Nozzle Geometry in Fused Filament Fabrication." *Additive Manufacturing* 28: 711–718. doi:10.1016/j.addma.2019.06.019.
- OpenFOAM, Free CFD Software, The OpenFOAM Foundation. (n.d.). Accessed 8 September 2020. <https://openfoam.org/>.
- Osswald, T. A., J. Puentes, and J. Kattinger. 2018. "Fused Filament Fabrication Melting Model." *Additive Manufacturing* 22: 51–59. doi:10.1016/j.addma.2018.04.030.
- Peng, F., B. D. Vogt, and M. Cakmak. 2018. "Complex Flow and Temperature History During Melt Extrusion in Material Extrusion Additive Manufacturing." *Additive Manufacturing* 22: 197–206. doi:10.1016/j.addma.2018.05.015.
- Phan, D. D., J. S. Horner, Z. R. Swain, A. N. Beris, and M. E. Mackay. 2020. "Computational Fluid Dynamics Simulation of the Melting Process in the Fused Filament Fabrication Additive Manufacturing Technique." *Additive Manufacturing* 33: 101161. doi:10.1016/j.addma.2020.101161.
- Phan, D. D., Z. R. Swain, and M. E. Mackay. 2018. "Rheological and Heat Transfer Effects in Fused Filament Fabrication." *Journal of Rheology* 62: 1097–1107. doi:10.1122/1.5022982.
- Pigeonneau, F., D. Xu, M. Vincent, and J. F. Agassant. 2020. "Heating and Flow Computations of an Amorphous Polymer in the Liquefier of a Material Extrusion 3D Printer." *Additive Manufacturing* 32, doi:10.1016/j.addma.2019.101001.
- Pimenta, F., and M. A. Alves. 2016. "RheoTool". Accessed 3 July 2020. <https://github.com/fppimenta/rheoTool>.
- Pimenta, F., and M. A. Alves. 2020. *RheoTool v5 User Guide*. <https://github.com/fppimenta/rheoTool>.
- QuickBooks. 2020. "RheoTool v5 UserGuide". Computer (Long. Beach. Calif). 1–20. www.impact-test.co.uk.
- Serdeczny, M. P., R. Comminal, M. T. Mollah, D. B. Pedersen, and J. Spangenberg. 2020b. "Numerical Modeling of the Polymer Flow Through the hot-end in Filament-Based Material Extrusion Additive Manufacturing." *Additive Manufacturing* 36: 101454. doi:10.1016/j.addma.2020.101454.
- Serdeczny, M. P., R. Comminal, D. B. Pedersen, and J. Spangenberg. 2020a. "Experimental and Analytical Study of the Polymer Melt Flow Through the hot-end in Material Extrusion Additive Manufacturing." *Additive Manufacturing* 32: 101454. doi:10.1016/j.addma.2019.100997.
- Van Doormaal, J. P., and G. D. Raithby. 1984. "Enhancements of the Simple Method for Predicting Incompressible Fluid Flows." *Numerical Heat Transfer* 7 (2): 147–163. doi:10.1080/01495728408961817.
- Wang, J., and D. F. James. 2011. "Lubricated Extensional Flow of Viscoelastic Fluids in a Convergent Microchannel." *Journal of Rheology* 55: 1103. doi:10.1122/1.3613948.
- Xia, H., J. Lu, S. Dabiri, and G. Tryggvason. 2018. "Fully Resolved Numerical Simulations of Fused Deposition Modeling. Part I: Fluid Flow." *Rapid Prototyping Journal* 24 (2): 463–476. doi:10.1108/RPJ-12-2016-0217.
- Xia, H., J. Lu, and G. Tryggvason. 2019. "A Numerical Study of the Effect of Viscoelastic Stresses in Fused Filament Fabrication." *Computer Methods in Applied Mechanics and Engineering* 346: 242–259. doi:10.1016/j.cma.2018.11.031.